

Syed Mohaiminul Hoque at AstroCLIMB 2026: Parameter-Efficient Adaptation of Qwen3-VL for Astronomy Citation Linking

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Abstract

We describe the solo submission of team *Syed Mohaiminul Hoque* to the ASTROCLIMB shared task at WASP 2026. The task requires classifying a pair of astronomy figures and/or captions as the same figure, from the same paper, from citation-related papers, or from unrelated papers. Our system casts classification as constrained next-token prediction with Qwen3-VL and adapts the model using 4-bit quantization and low-rank adapters. On the 10,000-pair Kaggle test set, three Qwen3-VL-8B models with language-attention adapters obtain 0.73230–0.73307 macro-F1 individually. Averaging their four-class probabilities raises the score to 0.73604, and masking the impossible same-figure class for caption–caption and image–image pairs produces 0.73725 and third place among 11 teams on the final leaderboard. This paper documents empirical system engineering rather than proposing a new learning algorithm; broader visual adaptation, restricted-loss blending, and citation-focused auxiliary training do not improve the result.

1 Introduction

Scientific figures condense results that are difficult to recover with text-only indexing. The ASTROCLIMB shared task asks systems to partially reconstruct an astronomy citation graph using only figures and captions (WASP 2026 Organizers, 2026). Unlike conventional document classification, the input is a pair of heterogeneous objects: either object may be an image or a caption, and the target is a symmetric relation between their source papers.

Modern vision–language models (VLMs) offer a natural interface for such mixed inputs. However, full fine-tuning remains expensive, especially on the two NVIDIA T4 GPUs available in a standard Kaggle environment. We therefore formulate

the task as instruction-following with four possible answer tokens and use QLoRA (Detrmers et al., 2023) to adapt Qwen3-VL (Bai et al., 2025a).

We develop a uniform multimodal pipeline with random object swapping; compare scale, loss normalization, adapter coverage, and seed variation; and analyze both a balanced validation split and a negative auxiliary-training result. Our three-seed probability ensemble with a modality constraint reaches 0.73725 test macro-F1, with the same_paper–related_papers distinction remaining the main difficulty. Code, notebooks, and artifacts are available at <https://github.com/SyedT1/Astroclimb-sharedtask>.

2 Related Work

Scientific-figure research commonly studies caption generation or cross-modal retrieval. SciCap introduced a large figure–caption corpus and captioning baselines (Hsu et al., 2021); SciMMIR benchmarks retrieval over 530K scientific image–text pairs (Wu et al., 2024). Chart retrieval also shows value in combining visual, textual, and derendered table evidence (Nowak et al., 2024). ASTROCLIMB instead predicts provenance relations rather than retrieving a matching caption.

SPECTER2 learns citation-informed scientific-document representations (Cohan et al., 2020; Singh et al., 2023); SigLIP2 and DINOv3 provide cross-modal and self-supervised visual features (Tschanen et al., 2025; Siméoni et al., 2025). We compare their frozen features with our VLM approach. Casting labels as answer tokens is related to pattern-based classification (Schick and Schütze, 2021); our contribution is a carefully evaluated shared-task system combining this interface with QLoRA, ensembling, and a task-valid constraint.

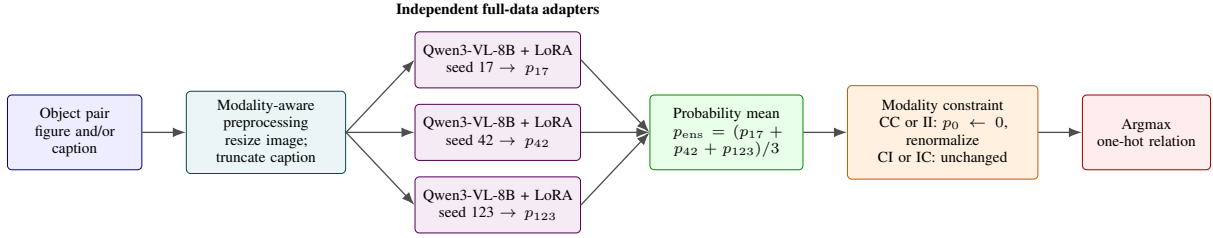


Figure 1: Inference pipeline of the best system. Three full-vocabulary Qwen3-VL-8B language-attention QLoRA models produce four-class probabilities. Equal averaging reduces seed variance, after which a deterministic modality mask removes the impossible same_figure option from caption–caption and image–image pairs.

3 Task and Data

Each example is a pair $x_i = (o_{i1}, o_{i2})$. An object is either an English figure caption or a base64-encoded scientific figure. The four mutually exclusive classes are: (0) same_figure, when the objects are the image and caption of one figure; (1) same_paper, when they are different objects from one paper; (2) related_papers, when one source paper cites the other; and (3) unrelated_papers. The relation is symmetric under exchanging the two objects.

The Kaggle release contains 10,000 labeled training pairs and 10,000 test pairs. Training labels are distributed (1000, 3000, 3000, 3000) in class order. The official metric is macro-averaged F1,

$$F_{1,\text{macro}} = \frac{1}{4} \sum_{c=0}^3 F_{1,c}, \quad (1)$$

so performance on the less frequent same_figure class contributes equally. The broader collection contains 94,233 figure–caption records with citation metadata (NASA Astrophysics Data System, 2026); we use it only for the auxiliary experiment in Section 5.4.

For controlled experiments, we reserve 200 examples per class using seed 42, producing a balanced 800-pair validation set and 9,200 training pairs. Full-data systems use all 10,000 labeled pairs. We do not use test metadata, external DOI lookup, or hidden labels.

4 System Description

4.1 Input representation

We use instruction-tuned Qwen3-VL-4B and Qwen3-VL-8B checkpoints. A system message defines the four relations and assigns them digits 0–3. The user message then presents Object 1 and Object 2 in their native modalities and asks for one

digit. Captions are limited to 3,000 characters by retaining both their beginning and end. Images are decoded to RGB and processed within an area range of 256^2 to 448^2 pixels. This cap is important for fitting the 8B model on T4 hardware.

Because the relation is symmetric, each training pair is randomly swapped with probability 0.5 without changing its label. We do not use swap test-time augmentation in the submitted systems.

4.2 QLoRA adaptation

We load the backbone in 4-bit NormalFloat (NF4) format with double quantization and FP16 computation. Low-rank adaptation (Hu et al., 2022) modifies a frozen weight W_0 as

$$W_{\text{eff}} = Q_{\text{NF4}}(W_0) + \frac{\alpha}{r} BA, \quad (2)$$

where $r = 16$, $\alpha = 32$, and LoRA dropout is 0.05. Our main systems adapt language-attention q/k/v/o projections; we also test adding visual-merger projectors or adapting all eligible linear layers.

Training uses two-process distributed data parallelism, one process per T4. The per-device batch size is one with eight gradient-accumulation steps, giving a global batch size of 16. The primary full-data systems train for one epoch with learning rate 10^{-4} . The 8B restricted-loss refit uses 5×10^{-5} for 1.5 epochs, selected on the fixed validation split.

4.3 Classification objectives

Let t_c be the tokenizer ID for digit c , and let $z_{i,v}$ be the logit immediately before the answer. The full-vocabulary objective supervises only that position:

$$\mathcal{L}_{\text{vocab}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp z_{i,t_{y_i}}}{\sum_{v \in \mathcal{V}} \exp z_{i,v}}. \quad (3)$$

We also test $\mathcal{L}_4$, replacing the denominator by the four valid digit tokens. Inference always selects

Backbone	Training setup	Epochs	Adapter coverage	Test macro-F1
Frozen encoders	SPECTER2/SigLIP2/DINO + features	—	modality CatBoost	0.48279
Qwen3-VL-4B	balanced 5K, $\mathcal{L}_{\text{vocab}}$	1.0	language attention	0.67856
Qwen3-VL-4B	full 10K, $\mathcal{L}_{\text{vocab}}$	1.0	language attention	0.70751
Qwen3-VL-4B	9.2K, $\mathcal{L}_4$	2.0	language attention	0.71453
Qwen3-VL-4B	full 10K, $\mathcal{L}_{\text{vocab}}$	1.0	language + visual projectors	0.71016
Qwen3-VL-4B	full 10K, $\mathcal{L}_{\text{vocab}}$	1.0	all linear layers	0.71202
Qwen3-VL-8B	full 10K, $\mathcal{L}_{\text{vocab}}$, seed 42	1.0	language attention	0.73230
Qwen3-VL-8B	full 10K, $\mathcal{L}_{\text{vocab}}$, seed 123	1.0	language attention	0.73250
Qwen3-VL-8B	full 10K, $\mathcal{L}_{\text{vocab}}$, seed 17	1.0	language attention	0.73307
Qwen3-VL-8B	full 10K, $\mathcal{L}_4$	1.5	language attention	0.73032
Qwen3-VL-8B	three-seed probability ensemble	1.0	language attention	0.73604
Qwen3-VL-8B	75% ensemble + 25% restricted	—	probability blend	0.73590
Qwen3-VL-8B	ensemble + modality mask	—	deterministic constraint	0.73725

Table 1: Final Kaggle leaderboard results. “Full 10K” denotes all labeled pairs; the 9.2K run retains 800 validation pairs. The final three rows are derived systems.

the largest of those four logits, avoiding unconstrained generation. Appendix A gives the full restricted-loss, inference, and swap definitions.

4.4 Probability ensemble and modality constraint

Figure 1 summarizes the best system. We independently train the full-vocabulary 8B configuration with seeds 17, 42, and 123. If $p_s(c | x)$ is the normalized four-class distribution from seed s , their equal-weight ensemble is

$$p_{\text{ens}}(c | x) = \frac{1}{3} \sum_{s \in \{17, 42, 123\}} p_s(c | x). \quad (4)$$

This is probability averaging rather than majority voting over hard predictions.

The same_figure label is valid only for a cross-modal figure–caption pair. For caption–caption (CC) and image–image (II) pairs, we impose the task constraint after ensembling:

$$\tilde{p}_0 = 0, \quad \tilde{p}_c = \frac{p_{\text{ens},c}}{\sum_{j=1}^3 p_{\text{ens},j}} \quad (c = 1, 2, 3). \quad (5)$$

Cross-modal probabilities remain unchanged. The mask modifies vectors for 6,000 of 10,000 test rows but changes only seven hard predictions. We do not use swap test-time augmentation, class-bias calibration, or the restricted-loss model in the final system.

5 Experiments

5.1 Zero-shot baselines

With the same prompt and 4-bit loading, Qwen3-VL-4B is the strongest zero-shot model at 0.47956 macro-F1; Qwen2.5-VL (Bai et al., 2025b) scores

0.46745. The frozen-encoder baseline in Table 1 reaches 0.48279; Appendix B details both baselines.

5.2 Fine-tuned systems

Table 1 summarizes the final-leaderboard results. Full-data 4B training raises macro-F1 from 0.47956 to 0.70751; the analogous 8B seed-42 model reaches 0.73230. Replications score 0.73250 and 0.73307, motivating probability averaging.

The best 8B model exceeds the comparable 4B model by 0.02556. Across three seeds, the mean is 0.73262 ± 0.00040 sample standard deviation (range 0.00077). Broader 4B adapter coverage adds only 0.00265–0.00451; a visual-projector-only run scores 0.48182 because caption–caption pairs bypass its trainable path.

The restricted-loss comparison also changes the split and epochs. Its selected 8B checkpoint obtains 0.729164 validation macro-F1; a full-data re-fit scores 0.73032. Blending 25% of its probabilities into the ensemble scores 0.73590, 0.00014 below the raw ensemble.

Probability averaging improves the best seed by 0.00297; the mask adds 0.00121, reaching 0.73725 and third of 11 teams. Seven predictions move from the impossible same_figure label to same_paper, with no extra inference.

5.3 Controlled validation analysis

On the balanced 800-pair split, the selected restricted-loss checkpoint obtains 0.72916 macro-F1. same_figure is easiest and related_papers weakest. Adding language MLP targets reaches 0.729019 while requiring more resources, supporting attention-only adaptation. Appendix B gives

the per-class results.

5.4 Citation-focused auxiliary training

We construct 20,000 auxiliary examples from the public AstroCLIMB collection (NASA Astrophysics Data System, 2026), distributed as (2000, 6000, 8000, 4000) in class order. Class 0 replays training examples with replacement; classes 1–3 pair captions from one DOI, directly citation-linked DOIs, or non-adjacent papers sharing a meaningful title token, with metadata used only for selection. UUID-pair keys enforce pair-level uniqueness for classes 1–3, and exact normalized-caption matches exclude recoverable validation and test DOIs, although image-only papers cannot be mapped. This auxiliary stage lowers macro-F1 from 0.72916 to 0.71405, likely because metadata relations need not be visible in the sampled captions; Appendix B gives detailed results.

6 Error Analysis

Three patterns emerge. First, exact figure-caption matching is comparatively tractable because shared entities, values, labels, and panel structure provide direct alignment. The raw ensemble nevertheless predicts `same_figure` for seven same-modality pairs, which the mask corrects. Second, same-paper and citation-related pairs often share topics, making similarity an unreliable proxy for provenance. Third, image-heavy inputs are harder than text-heavy inputs; modality-level diagnostics are reported in Appendix B.

The result that language-side adaptation outperforms broad visual adaptation is not evidence that figures are unimportant. Instead, 10,000 labels may be insufficient to tune many visual parameters, while the pretrained visual encoder already supplies useful features. Better figure-specific pre-training, OCR, and evidence-aware sampling are promising directions.

7 Efficiency and Reproducibility

The restricted 8B run trains 15,335,424 parameters for 938 steps in 320.45 minutes, peaking at 11.53 GiB on rank 0. Two-T4 inference over 10,000 pairs takes 78.15 minutes. The final system requires three independently trained and inferred adapters, but ensembling and masking are CPU-only. We use Transformers 4.57.1, PEFT 0.17.1, Accelerate 1.10.1, and bitsandbytes 0.47.0. The split seed is 42; full-data seeds are 17, 42, and

123. Appendix C reports the optimizer, scheduler, checkpoint policy, and output validation.

8 Conclusion

Our language-attention QLoRA system constrains Qwen3-VL prediction to four tokens. One 8B model reaches 0.73307 test macro-F1, three-seed averaging reaches 0.73604, and the modality mask yields 0.73725, ranking third of 11 teams. Broader visual adaptation, restricted-loss blending, and citation-graph supervision do not improve the result. Future work should prioritize evidence-visible auxiliary examples and stronger cross-figure textual grounding.

Limitations

Most full-data configurations lack local validation, and differences are measured by single final-leaderboard scores. Final labels and example-level predictions are unavailable, so bootstrap confidence intervals and paired significance tests cannot be computed; the reported seed standard deviation measures run variability, not a confidence interval. The controlled split uses one seed and may share source papers across partitions because paper identifiers are unavailable. Some comparisons vary multiple factors, including data size and epoch count. Caption truncation and image resizing can remove evidence. Finally, metadata-defined citation relations may be invisible in the presented objects, imposing a task-level ceiling that we do not estimate.

Ethics Statement

The system processes public scientific figures and captions supplied for the shared task. It is a benchmark model, not an authoritative bibliographic database: predicted links can be wrong and should not assign credit, evaluate researchers, or alter citation records without verification. We use only released data and public task metadata and infer no sensitive personal attributes. GPU computation is the main environmental cost; 4-bit quantization, LoRA, and constrained one-step inference reduce it relative to full-model training and free-form generation.

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A Objective and Prompt Details

Because the ground-truth relation is symmetric, each training pair is swapped with probability 0.5:

$$(\tilde{o}_{i1}, \tilde{o}_{i2}) = \begin{cases} (o_{i2}, o_{i1}) & b_i = 1, \\ (o_{i1}, o_{i2}) & b_i = 0, \end{cases} \quad (6)$$

$$b_i \sim \text{Bernoulli}(0.5).$$

The label is unchanged. For the restricted objective, the denominator contains only the four valid digit tokens:

$$\mathcal{L}_4 = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp z_{i,t_{y_i}}}{\sum_{c=0}^3 \exp z_{i,t_c}}. \quad (7)$$

Regardless of the training objective, inference reads only those digit logits:

$$\hat{y}_i = \arg \max_{c \in \{0,1,2,3\}} z_{i,t_c}. \quad (8)$$

The system instruction used in the main runs is:

You classify the relationship between two objects from astronomy papers.
0: The objects are the figure and caption of the same scientific figure.
1: The objects are from different figures in the same paper.
2: The objects are from different papers and one paper cites the other.
3: The objects are from unrelated papers.
The relationship is symmetric. Output only one digit: 0, 1, 2, or 3.

Each object is introduced as either a “scientific figure” or a “figure caption.” The final user instruction is: “Classify their relationship. Reply with one digit only.”

B Additional Experimental Results

Our non-generative baseline combines frozen SPECTER2 caption embeddings (Singh et al., 2023), SigLIP2 image–text embeddings (Tschanen et al., 2025), and DINO visual embeddings (Siméoni et al., 2025) with word/character TF-IDF, OCR overlap, and perceptual-hash similarities. Three modality-specific CatBoost classifiers handle caption–caption, mixed, and image–image pairs. It is therefore a frozen-feature classifier pipeline, not a SPECTER-only model. Its submission scores 0.48279 final-leaderboard macro-F1.

The auxiliary stage contains 20,000 pairs distributed as (2000, 6000, 8000, 4000), followed by restricted-loss tuning on 9,200 competition

Model	Test macro-F1
Qwen3-VL-4B-Instruct	0.47956
Qwen2.5-VL-7B-Instruct	0.46745
Qwen3-VL-8B-Instruct	0.40112

Table 2: Zero-shot results on the 10,000-pair test set.

Metric	F1
same_figure	0.92913
same_paper	0.66988
related_papers	0.53790
unrelated_papers	0.77975
Macro average	0.72916

Table 3: Fixed-split results for restricted-loss 8B QLoRA.

pairs. Its confusion matrix (gold rows, predicted columns) is

$$\begin{bmatrix} 165 & 30 & 5 & 0 \\ 4 & 131 & 62 & 3 \\ 0 & 51 & 113 & 36 \\ 0 & 5 & 38 & 157 \end{bmatrix}. \quad (9)$$

The largest confusion is bidirectional between same_paper and related_papers (62 and 51 cases). Auxiliary training improves the two different-paper classes but degrades same_figure and same_paper. Modality-level macro-F1 is 0.72845 on caption–caption, 0.67108 on caption–image, and 0.65102 on image–image pairs. These values diagnose the auxiliary model, not the full-data submission.

C Primary Configuration and Reproducibility

The full-data seed models use the final one-epoch adapter without validation-based selection; intermediate recovery checkpoints are saved every 100 steps. The restricted validation experiment evaluates at quarter-run milestones and selects checkpoint 863 (1.5 epochs) by maximum validation macro-F1 before the locked full-data refit. The restricted 8B run trains 15,335,424 parameters for 938 optimizer steps in 320.45 minutes, with 11.53 GiB peak allocated memory on rank 0. Two-GPU inference over 10,000 pairs takes 78.15 minutes. Combining stored probabilities and applying the modality mask require no model inference. The implementation uses Transformers 4.57.1, PEFT 0.17.1, Accelerate 1.10.1, and bitsandbytes 0.47.0, and validates unique IDs, exact

Metric	Baseline	+Auxiliary
Macro-F1	0.72916	0.71405
same_figure	0.92913	0.89431
same_paper	0.66988	0.62830
related_papers	0.53790	0.54067
unrelated_papers	0.77975	0.79293

Table 4: Per-class F1 before and after auxiliary training.

Setting	Value
Backbone	Qwen3-VL-8B-Instruct
Component seeds	17, 42, 123
Training pairs	10,000 per seed
Epochs	1 per seed
Learning rate	10^{-4}
Optimizer	paged AdamW, 8-bit
Scheduler	cosine; 5% warm-up
Weight decay	0.01
Gradient clipping	maximum norm 1.0
Quantization	4-bit NF4, double quantization
Compute dtype	FP16
LoRA (r, α, p)	(16, 32, 0.05)
LoRA targets	language $q/k/v/o$ projections
Global batch size	16
Maximum image area	448×448
Maximum caption length	3,000 characters
Training augmentation	object swap, $p = 0.5$
Ensemble	equal mean of three probability vectors
Postprocessing	mask class 0 for CC and II pairs
Inference	four digit logits, ensemble, mask, argmax
Hardware	$2 \times$ NVIDIA T4; CPU
Checkpoint policy	postprocessing final epoch; no validation or early stopping

Table 5: Configuration of the best leaderboard system.

column order, finite normalized probabilities, binary targets, one prediction per row, and 10,000 output rows.